PES UNIVERSITY

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Department of Computer Science and Engineering

UE24MA241B – Linear Algebra and Its Applications

UNIT 1

1. Solving Linear Systems & LU Decomposition

a. **np.array()**
   - Used to define matrices (like the coefficient matrix A) and vectors (like the constant vector b) as numerical arrays for computation.

b. **scipy.linalg.lu(a)**
   - Performs LU decomposition on a matrix. It returns three matrices:
     - **P (Permutation matrix):** Accounts for any row swaps needed during decomposition.
     - **L (Lower triangular matrix):** Contains the multipliers used during Gaussian elimination.
     - **U (Upper triangular matrix):** The final row-echelon form of the matrix.

c. **np.linalg.solve(a, b)**
   - A high-performance function used to solve a linear matrix equation Ax = b.
   - Used as the standard for LU-based solving.

d. **np.dot(a, b)**
   - Computes the dot product of two arrays.
   - Used to verify the solution by calculating Ax and comparing it to b to find the residual error.

e. **np.linalg.norm(x)**
   - Computes the magnitude (L2 norm) of a vector.
   - Used to measure the residual error (||Ax - b||) to check the accuracy of the solution.

2. Computational Efficiency & Matrix Operations

a. **np.eye(n)**
   - Generates an identity matrix of size n × n.
   - Used to ensure the random matrix A is diagonally dominant and invertible.

b. **np.random.rand(m, n)**
   - Generates an m × n matrix with random values in the range [0,1).

- Used to create test matrices for efficiency comparison.

c. **np.linalg.inv(a)**
   - Computes the inverse of a matrix.
   - Used to compare solving using $x = A^{-1}b$ versus direct decomposition.

d. **@ (Matrix Multiplication Operator)**
   - Standard Python operator for matrix multiplication.
   - Used to multiply the inverse of A with b.

e. **np.hstack(tup)**
   - Stacks arrays horizontally (column-wise).
   - Used to create an augmented matrix [A | b] for RREF calculation.

f. **sp.Matrix(c).rref()**
   - Function from SymPy that computes the Reduced Row Echelon Form (RREF).
   - Used to demonstrate the computational cost of symbolic row reduction.

g. **time.time()**
   - Returns the current time in seconds.
   - Used to measure execution time of different methods (LU, inverse, RREF).

3. Sports Ranking (Colley's Method)

a. **scipy.io.loadmat(filename)**
   - Loads MATLAB .mat files into Python.
   - Used to import the scores dataset.

b. **np.diag(v)**
   - Extracts or constructs a diagonal matrix.
   - Used to build the Colley Matrix with total games played on the diagonal.

c. **np.sum(a, axis=1)**
   - Sums elements across rows.
   - Used to calculate:
     - Total games played
     - Net score (wins vs losses) for vector b

d. **np.abs(a)**
   - Computes the absolute value element-wise.
   - Used to determine total number of games played from the score matrix.

e. **np.argsort(a)[::-1]**
   - **argsort:** Returns indices that would sort an array
   - **[::-1]:** Reverses order
     - Used to rank teams from highest to lowest Colley rating.